Motion in Two Dimensions:

- ❖ motion in a plane (x and y Coordinate). Here, the direction of the force acting on an object will not be in the same line as its initial velocity. Thus, the velocity and acceleration will have different directions. For this reason we have to use vector equations.

1. Average and Instantaneous Velocities: The definitions of displacement, average and instantaneous velocities, average and instantaneous speeds v and acceleration will be the same as those for rectilinear motion except that each of these quantities will now have components along the *x* and *y* directions.

a. **The position** of the object will be described by its position vector $\vec{r_1}$ & $\vec{r_2}$ are:

$$\vec{r_1} = x_1\hat{\imath} + y_1\hat{\jmath} \qquad \& \qquad \vec{r_2} = x_2\hat{\imath} + y_2\hat{\jmath}$$

b. **The displacement** of the particle from t_2 to t_1 i.e., in time $t = t_2 - t_1$ is given by

$$\overrightarrow{\Delta r} = \vec{r_2} - \vec{r_1} = (x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{\jmath}$$

c. **The average velocity** of the object is a vector whose direction is along $\overrightarrow{\Delta r}$ and given by:

$$\overrightarrow{V_{avg}} = \frac{\Delta\vec{x}}{\Delta t} = \left(\frac{x_2 - x_1}{t_2 - t_1}\right)\hat{\imath} + \left(\frac{y_2 - y_1}{t_2 - t_1}\right)\hat{\jmath}$$

Where: $$\left(\mathrm{v_{avg}}\right)_{\mathrm{x}} = \left(\frac{x_2 - x_1}{t_2 - t_1}\right) \;\&\left(\mathrm{v_{avg}}\right)_{\mathrm{y}} = \left(\frac{y_2 - y_1}{t_2 - t_1}\right)$$

Thus: $$\overrightarrow{\mathrm{V_{avg}}} = \left(\mathrm{v_{avg}}\right)_{\mathrm{x}}\hat{\imath} + \left(\mathrm{v_{avg}}\right)_{\mathrm{y}}\hat{\jmath}$$

The **magnitude and direction** of the instantaneous velocity are given by:

$$\mathrm{v_{avg}} = \sqrt{\left(\mathrm{v_{avg}}\right)_{\mathrm{x}}^2 + \left(\mathrm{v_{avg}}\right)_{\mathrm{y}}^2} \qquad \& \qquad \tan\theta = \frac{\left(\mathrm{v_{avg}}\right)_{\mathrm{y}}}{\left(\mathrm{v_{avg}}\right)_{\mathrm{x}}}$$

d. The **instantaneous velocity** of the object can be written in terms of derivative as

$$\vec{V} = \lim_{\Delta t \to 0}\left(\frac{\Delta\vec{r}}{\Delta t}\right) = \left(\frac{d\vec{x}}{at}\right)\hat{\imath} + \left(\frac{d\vec{y}}{at}\right)\hat{\jmath}$$

The magnitude and direction of the instantaneous velocity are given by:

$$v = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

2. Average and Instantaneous Acceleration:

acceleration: rate of change of velocity where **instantaneous acceleration** is acceleration at a particular instant, and **Average Acceleration**: change of velocity divided by the time interval for that change.

a. Average Acceleration ($\vec{\boldsymbol{a}}_{\boldsymbol{av}}$) of a particle between times (t_2 and t_1) can be written as:

$\vec{a}_{av} = \frac{v_2 - v_1}{t_2 - t_1} = \left[\frac{v_{2x} - v_{1x}}{t_2 - t_1}\right]\hat{\imath} + \left[\frac{v_{2y} - v_{1y}}{t_2 - t_1}\right]\hat{\jmath}$

where (v_2 and v_1) are the velocities of the particle at times (t_2 and t_1) respectively.

$$\vec{a}_{av} = (a_{av})_x\hat{\imath} + (a_{av})_y\hat{\jmath}$$

$(\vec{a}_{av})_x$ *and* $(\vec{a}_{av})_y$ being the x and y components of the average acceleration

b. The magnitude and direction of the acceleration are given by:

$a_{av} = \sqrt{(a_{av})_x^2 + (a_{av})_y^2}$ and $\tan\theta = \frac{(a_{av})_y}{(a_{av})_x}$

b. Average and Instantaneous Acceleration (Cont.):

c. The instantaneous acceleration is given by

$$\vec{a} = \lim_{\Delta l \to 0}\left[\frac{\Delta v}{\Delta t}\right] = \frac{dv}{dt} = \left[\frac{dv_x}{dt}\right]\hat{\imath} + \left[\frac{dv_y}{dt}\right]\hat{\jmath}$$

$$= \frac{d}{d\vec{t}}\left[\frac{dx}{dt}\right]\hat{\imath} + \frac{d}{d\vec{t}}\left[\frac{dy}{dt}\right]\hat{\jmath}$$

$$= \left[\frac{d^2x}{dt^2}\right]\hat{\imath} + \left[\frac{d^2y}{dt^2}\right]\hat{\jmath}$$

d. The magnitude and direction of the instantaneous acceleration are given by:

$a = \sqrt{\left[\frac{d^2\mathrm{x}}{\mathrm{dt}^2}\right]^2 + \left[\frac{d^2y}{\mathrm{dt}^2}\right]^2}$ and $\tan\theta = \frac{dv_\mathrm{y}}{dt} / \frac{dv_\mathrm{x}}{dt} = \frac{dv_\mathrm{y}}{dv_\mathrm{x}}$

which is the slope of the tangent to the curve in velocity graph, i.e., a plot of (v_y) versus (v_x)

c. The relative velocity: The relative velocity of object A having velocity $\vec{v}_\mathrm{A}$, with respect to the object B having velocity $\vec{v}_\mathrm{B}$, is given by: $\vec{v}_\mathrm{AB} = \vec{v}_\mathrm{A} - \vec{v}_\mathrm{B}$

Similarly, the relative velocity of object B with respect to object A, is given by: $\vec{v}_\mathrm{BA} = \vec{v}_\mathrm{B} - \vec{v}_\mathrm{A}$

d. Projectile Motion:

In this motion, we have objects (projectiles) with given initial velocity, moving under the influence of the Earth's gravitational field. The projectile has two components of velocity, one in the horizontal, i.e., along x-direction and the other in the vertical, i.e., along the y direction. The acceleration due to gravity acts only along the vertically downward direction. The horizontal component of velocity, therefore, remains unchanged as no force is acting in the horizontal direction, while the vertical component changes in accordance with laws of motion with $\vec{a}_\mathrm{x}$ being 0 and $\vec{a}_\mathrm{y}$ (= $-\vec{g}$) being the downward acceleration due to gravity (upward is positive). Unless stated otherwise, retarding forces like air resistance, etc., are neglected for the projectile motion.

e. Uniform Circular Motion (UAM): An object moving with constant speed along a circular path. In order to realize a circular motion, there are two requirements; (i) **tangential velocity** and (ii) **centripetal force o**f suitable **constant magnitude**.

- **Period, Radius Vector and Angular Speed:**

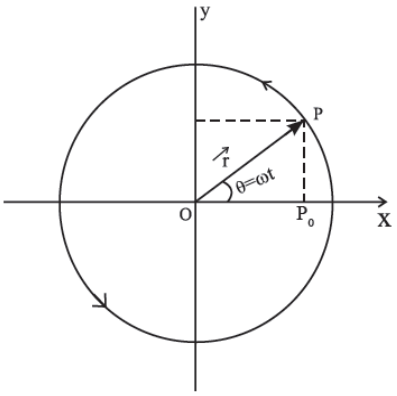
Consider an object of mass *m*, moving with a uniform speed v, along a circle of radius *r*. Let *T* be the time period of revolution of the object, i.e., the time taken by the object to complete one revolution or to travel a distance of 2π*r*. Thus,

$$speed\ v = \frac{Distance}{Time} = \frac{2\pi r}{T} \qquad \& \quad T = \frac{2\pi r}{v}$$

During one complete revolution, the angle described is 2π and the time taken is period T. Hence, the angular speed:

$$\omega = \frac{Angle}{time} = \frac{2\pi}{T} = \frac{(2\pi)}{\left(\frac{2\pi r}{v}\right)} = \frac{v}{r}$$

e. Uniform Circular Motion (UAM) (Cont.):



The radius vector of P makes an angle θ with the *x*-axis.

x and *y* components of the radius vector (*r)*

will then be (*r* cosθ) and (*r* sinθ) respectively.

$$\vec{r} = (r\cos\theta)\hat{\imath} + (r\sin\theta)\hat{\jmath}$$

$$= (r\cos[\omega t])\hat{\imath} + (r\sin[\omega t])\hat{\jmath}$$

Time derivative of position vector (*r)* gives instantaneous velocity (v),

$$\vec{V} = \left(\frac{d\vec{r}}{at}\right) = r(-\omega\, sin[\omega t])\hat{\imath} + (\omega\cos[\omega t])\hat{\jmath}$$

$$= r\,\omega(-\sin[\omega t])\hat{\imath} + \cos[\omega t])\hat{\jmath}$$

and time derivative of velocity (v) gives instantaneous acceleration (*a):*

$$\vec{a} = \left(\frac{d\vec{v}}{at}\right) = r\omega(-\omega cos[\omega t])\hat{\imath} - (\omega \sin[\omega t])\hat{\jmath}$$

$$= -\boldsymbol{\omega^2}(rcos[\omega t])\hat{\imath} + r\sin[\omega t])\hat{\jmath} = \boldsymbol{\omega^2}\vec{r}$$

e. Uniform Circular Motion (UAM) (Cont.):

Magnitudes of Acceleration:

$$a = \omega^2 r = \frac{v^2}{r^2} * r = \frac{v^2}{r} = \omega v$$

- The force providing this acceleration should also be along the same direction, hence centripetal.

  $$\vec{F} = m\vec{a} = -m\omega^2\vec{r}$$

- Magnitude of F:

  $$F = m\omega^2 r = m\omega v$$

Energy

- ❖ Every physical process that occurs in the Universe involves energy and energy transfers or transformations.
- ❖ **Systems**

A system is a small portion of the Universe.

- ❖ **A valid system** :May be a single object or particle. may be a collection of objects or particles, may be a region of space, may vary with time in size and shape

System Example

· A force applied to an object in empty space.

Energy: is a conserved quantity, a property of objects and systems that is quantified by the capacity to perform work. Is the ability to do work or cause change. The SI unit for energy is the joule (J).
You need energy when To move a box, To light a bulb, To heat your home.

Energy

Common Forms of Energy: Energy categorized as either **kinetic** or **Potential**.

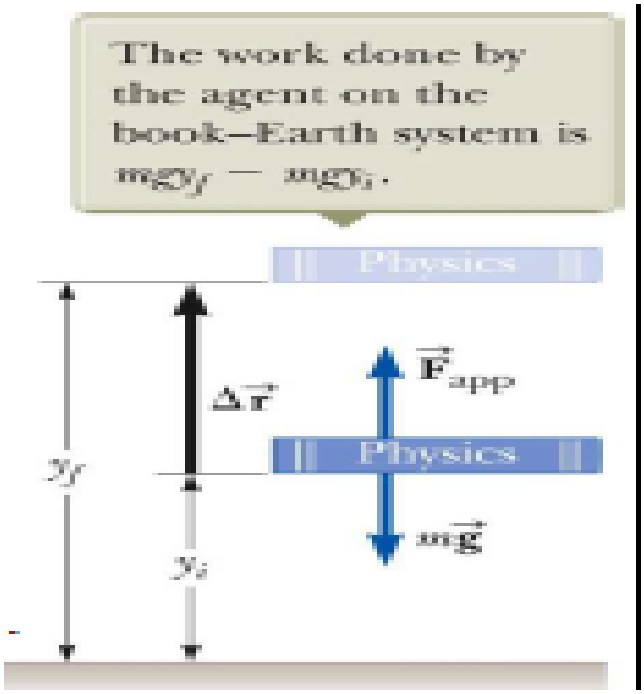
1. **Potential Energy (Stored Energy):**
   The system is the Earth and the book. Do work on the book by lifting it slowly through a vertical displacement $\Delta\vec{r} = (y_f - y_i)\hat{\jmath}$

2. The work done on the system must appear as an increase in the energy of the system.It is energy storage mechani . This is the energy stored in an object due to its position, state, or configuration. Such as:

a. Chemical Potential Energy: Energy stored in the bonds of atoms and molecules. (e.g., food, gasoline, batteries).
b. **Elastic Potential Energy:** Energy stored in stretched or compressed objects. (e.g., a drawn bowstring, a compressed spring).
C. **Nuclear Potential Energy:** Energy stored in the nucleus of an atom, holding protons and neutrons together. This is released in nuclear fission and fusion (e.g., in nuclear power plants or the sun).

Energy

d. Gravitational Potential Energy: Is the energy associated with an object at a given location above the surface of the Earth**.** Energy stored due to height. (e.g., water at the top of a dam, a book on a high shelf).



$$W_{ext} = (\bar{F}_{app}) \cdot \Delta\bar{r}$$

$$= (mg\hat{j}) \cdot [(y_f - y_i)]\hat{j} = mgy_f - mgy_i$$

❖ The quantity mgy is identified as the gravitational potential energy

$$U_g = mgy$$

U_g is a scalar. Units are joules (J), 1 Joule = 1 kg·m^2/s^2, m: mass, g: gravitational acceleration, h: height.

- Work may change the gravitational potential energy of the system.

$$W_{ext} = \Delta U_g$$

U= mgh SI Unit: Joules (J).

- **Example in life sciences:**
  - Blood raised in arteries (by the heart) gains potential energy that drives circulation back down.
  - A lifted limb has gravitational potential energy; when it falls, that energy converts into motion or heat.

Energy

- **2. Kinetic Energy (Energy of Motion) (K):**
  One possible result of work acting as an influence on a system is that the system changes its speed. The system could possess kinetic energy.

- Kinetic Energy is the energy of a particle This is the energy of an object possesses due to its motion. e.g. A rolling ball, flowing water in a river, the vibration of atoms (which we feel as heat), visible light, and sound waves.

$$K = \frac{1}{2}mv^2$$ (where m is mass (kg) and v is speed (m/s)).

SI Units: Joules (J). 1 Joule = 1 kg·m^2/s^2

- Properties:

· It is a scalar quantity (it has magnitude, no direction).

· It is always zero or positive (it depends on v^2).

· It depends on the reference frame (your speed depends on who is measuring it).

Work

- **Work:** Is the process of transferring energy to an object by applying a constant force that causes the object to move.

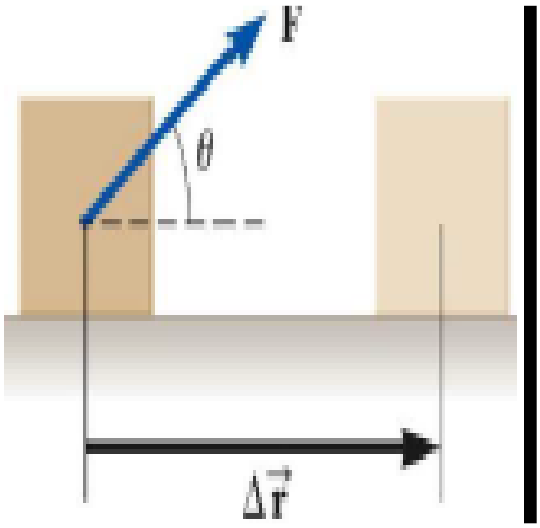
- It has done on a system by an agent exerting a constant force on the system is the product of the magnitude F of the force. the magnitude (Δr) of the displacement of the point of application of the force, and $(\cos\theta)$,where $(\theta)$is the angle between the force and the displacement vectors.

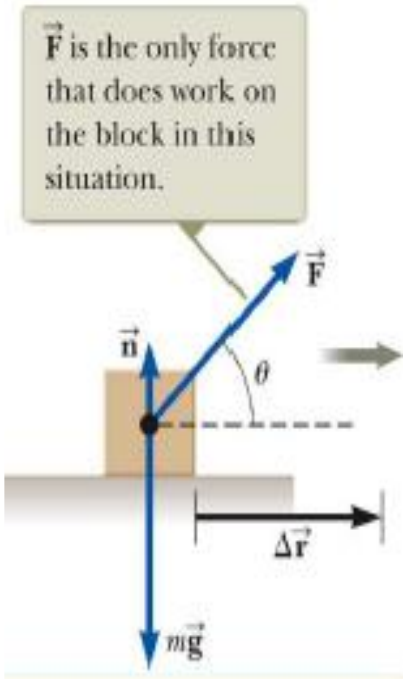
$$W = F \cdot \Delta r = F\Delta r \cos\theta$$

F: is the magnitude of the constant force.

Δr : is the magnitude of the displacement vector.

θ: the same as energy. is the angle between the force vector and the displacement vector.





Work

Work is a scalar quantity.

SI Units: Joules (J) =N.m

1 Joule = 1 newton .1meter = kg · m^2 / s^2

A force does no work on the object if the force does not move through a displacement.

- The work done by a force on a moving object is zero when the force applied is perpendicular to the displacement of its point of application.
- The normal force and the gravitational force do no work on the object when it moves horizontally.

$\cos\theta = \cos 90^0 = 0$

- $F\Delta r \cos\theta$: is the component of the force in the direction of the displacement. Only this component does work.

Work

- The angle ϕ is crucial and leads to three distinct cases:

➢ Work is **POSITIVE** (ϕ < 90°): The force has a component in the same direction as the displacement. It "helps" the motion and increases the object's kinetic energy.

Example: The force of gravity on a falling object (ϕ = 0°, cos0° = 1).

- Work is **ZERO** ($\phi = 90°$): The force is perpendicular to the displacement. It does not affect the object's speed.

Example: The centripetal force in uniform circular motion.

Example: The normal force on an object sliding horizontally.

- Work is **NEGATIVE** ($\phi > 90°$): The force has a component opposite to the displacement. It "hinders" the motion and decreases the object's kinetic energy.

Example: The force of kinetic friction ($\phi = 180°$, $\cos 180° = -1$).

Work

- **Biological examples:**
  - Muscles do work when lifting or moving objects converting chemical energy (ATP) into mechanical energy (movement).

(ATP Adenosine triphosphate is it produce in the mitochondria, which are knows as the " powerhouses " of the cells. Is the main energy molecule in the human body. It provide energy for muscle, heart, the lungs, the brain, and all body functions. When cell needs energy, it breaks down ATP to released that energy) it is important for moving muscles, pump blood, breathe, build or repair cells, send nerve signal).

  - The heart does work to pump blood against pressure.

The heart performs work when it pumps blood. It pushes blood against the pressure inside the blood vessels, which requires energy.

Work

- **Biological examples:**
  - The lungs do work during breathing expanding and compressing air.

The lungs perform work while inhaling and exhaling.
During inhalation, the lungs expand to take in air.
During exhalation, they contract to

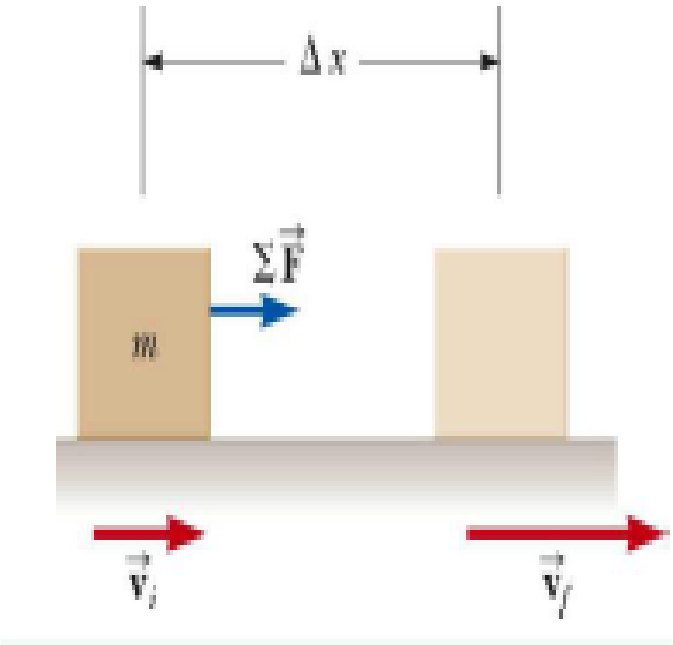
- Work is the mechanism by which energy is transferred between systems.

The Work-Kinetic Energy Theorem:

- A change in kinetic energy is one possible result of doing work to transfer energy into a system .

The net work done on a particle by all the forces acting on it is equal to the change in its kinetic energy."

$$W_{net} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$



- It directly relates the total work done over a distance to the change in speed, without needing to know the acceleration or the time involved.

- **Work Done by a Variable Force (in One Dimension)**

Forces are not always constant. To calculate the work done by a variable force,

we use calculus.

The Concept: Break the displacement into many tiny segments dx. Over each tiny

segment, the force F(x) is approximately constant. The work done in that segment is:

$$dW = F(x)\,dx\cdot$$

The total work is the sum (integral) of all these infinitesimal works:

$$W = \int_{x=i}^{x=f} F(x)\,dx = \int_{x=i}^{x=f} ma\,dx$$

Work

- **Example:**

  Work Done by a Spring:

The spring is the most important example of a variable force obeying Hooke's Law.

Hooke's Law: The force exerted by a spring is:

$$F_s = -\mathrm{Kx}$$

k: is the spring constant (stiffness) in N/m, x: is the displacement from the spring's relaxed (equilibrium) position. The negative sign indicates the spring force is a restoring force; it always acts to bring the spring back to equilibrium.

Work

- **Example (Conts):**

  Work Done by a Spring:

Work Done by an Applied Force to Stretch/Compress a Spring: To stretch a spring slowly (with no change in kinetic energy), you must apply a force equal and opposite to the spring's force:

$$F_{applied} = +\mathrm{Kx}$$

$$\mathrm{W} = \int_0^x F_{applied}\, dx = \int_0^x Kx\, dx = \frac{1}{2}\mathrm{K}x^2$$

Work Done by the Spring Force: Since the spring force is: $F_s = -Kx$

The work done by the spring is:

$$W_s = \int(-Kx)\, dx = -\frac{1}{2}Kx^2$$

Power

- **Power**: The rate at which work is done or energy is transferred.
- **Average Power:** $P_{avg} = \frac{\Delta W}{\Delta t}$
- **Instantaneous Power**: $\mathrm{P} = \frac{d\mathrm{W}}{\mathrm{dt}}$
- For a constant force, this can be related to velocity: $\mathrm{P} = \frac{d(Fx)}{\mathrm{dt}} \rightarrow \mathrm{P} = F\frac{\mathrm{dx}}{\mathrm{d}t} = Fv$
- **Units:** The SI unit of power is the Watt (W). & $(1\ W = 1\ Js)$.

Power

- **Biological Context**:
  - The metabolic rate of an organism is a measure of the power it generates (or consumes). For example, the power output of a muscle can be calculated when is contract.
- **In living systems:**

- The human heart at rest delivers (≈ 1.3 W). This is sufficient to pump blood throughout the body.
- During exercise, total body power output can exceed 1,000 W (1 kilowatt). This is the combined power from all the working muscles.
- The Natural Limit: There is a maximum amount of mechanical power that muscles can produce. This limit is dictated by the metabolic rate that is, the rate at which the body can convert energy from food (with oxygen) into useful work.
- The metabolic rate (from food energy) limits how much mechanical power muscles can produce.

❖ **Types of Systems**

Non-isolated systems : Energy can cross the system boundary in a variety of ways.

Total energy of the system changes.

Isolated systems : Energy does not cross the boundary of the system.

Total energy of the system is constant.

❖ **Conservation of energy**

Can be used if no non-conservative forces act within the isolated system.

Applies to biological organisms ,technological systems, engineering situations, etc.

❖ **Ways to Transfer Energy Into or Out of A System**

In non-isolated systems, energy crosses the boundary of the system during some time interval due to an interaction with the environment.

Work :transfers energy by applying a force and causing a displacement of the point of application of the force.

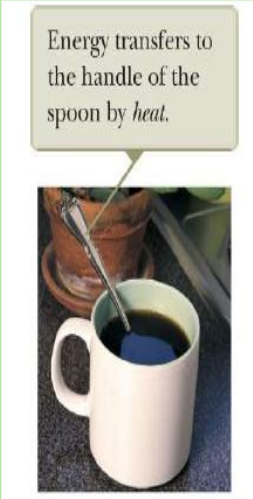
Mechanical Wave : Transfers energy by allowing a disturbance to propagate through a medium.

Heat : The mechanism of energy transfer that is driven by a temperature difference between two regions in space.

Matter Transfer : Matter physically crosses the boundary of the system, carrying energy with it.

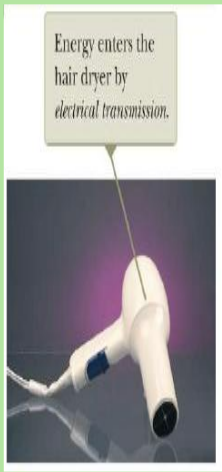
Electrical Transmission : Energy transfer into or out of a system by electric current.

Electromagnetic Radiation : Energy is transferred by electromagnetic waves



Energy transfers to the handle of the spoon by heat.
Explanation: This describes thermal conduction.
Mechanism: When a spoon is placed in a hot medium (like soup), heat energy transfers from the hotter end (in the soup) to the cooler handle. This occurs through the vibration and collision of atoms or molecules within the spoon's material.
· Example: The handle of a metal spoon becomes warm after being

Energy leaves the light-bulb by electromagnetic radiation.
(Correction: "light-buffs" → "light-bulb")
Explanation: This describes energy leaving a system as electromagnetic waves (light and heat).
Mechanism: A light bulb converts electrical energy into light (visible electromagnetic radiation) and infrared radiation (heat). This energy travels through space as waves.
· Example: An incandescent



Energy enters the hair dryer by electrical transmission.
(Correction: "fuel dryer" → "hair dryer", and "thermal reexamination" → "electrical transmission")
Explanation: This describes energy entering a device via an electric current.
Mechanism: Electrical energy is delivered through wires from a power source (a wall outlet) to the hair dryer. This electrical energy is then converted into

Energy leaves the radio from the speaker by sound waves.
(Correction: "mechanical source" → "sound waves" or "mechanical waves")
Explanation: This describes energy leaving a system as sound, a type of mechanical wave.
Mechanism: The radio converts electrical signals into mechanical

Energy enters the automobile gas tank by matter transfer.
Explanation: This describes energy entering a system by transferring matter that contains stored energy.
Mechanism: Gasoline (or any fuel) contains chemical potential energy within its molecular bonds. When fuel is pumped into the tank, you are physically transferring this energy-carrying matter into the car's

left in a hot cup of tea.	bulb glowing and emitting light and warmth.	other forms: thermal energy (heating element) and mechanical energy (fan motor). · Example: Plugging in and turning on any electrical appliance.	vibrations in the speaker cone. These vibrations create pressure waves in the air, which we perceive as sound. Sound waves carry energy away from the speaker. Example: Music or speech being projected from a speaker.	system. Example: Filling up a car's fuel tank at a gas station.

Conservation of Mechanical Energy

- **The Principle of Conservation of Mechanical Energy:**

Mechanical Energy (E): The sum of kinetic and potential energies.

$$E = k + u$$

- **The Law:** If only conservative forces do work within an isolated system, the total mechanical energy of the system is constant ($E_i = E_f$):

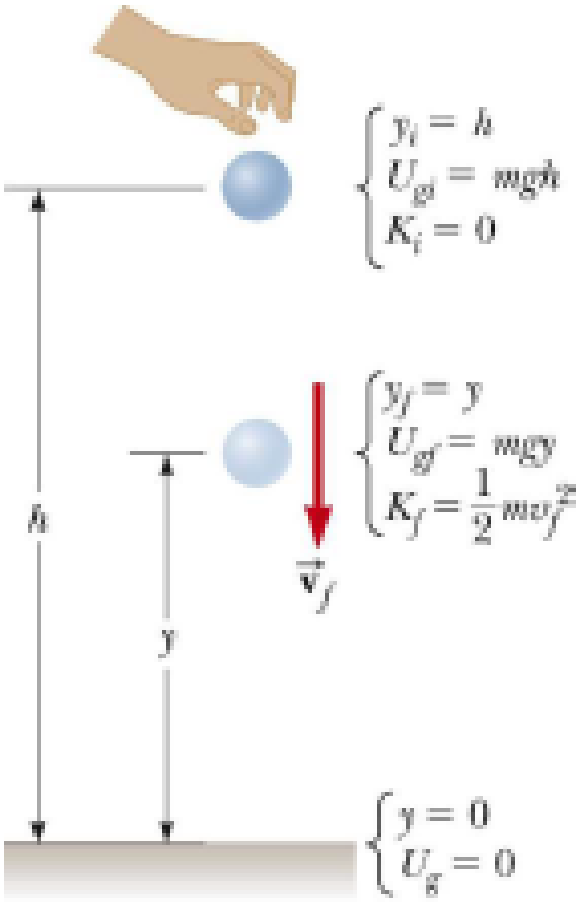
$$K_i + U_i = K_f + U_f$$

- What it means: Energy can be transformed from kinetic to potential and back again, but the total amount remains unchanged.

Example – **Ball in Free Fall**

Determine the speed of the ball at a height y above the ground .

Conceptualize: Use energy instead of motion.



Categorize:

System is the ball and the Earth.

System is isolated. Use the isolated system model.

Only force is gravitational which is conservative.

Analyze:

Apply the Conservation of Mechanical Energy.

$$K_i + U_i = K_f + U_f$$

$K_i = 0$, the ball is dropped.

Solve for V_f ,$V_f = \sqrt{2g(h-y)}$

Finalize :The equation for V_f is consistent with the results obtained from the particle under constant acceleration model for a falling object.

(Free Fall): As a ball falls, its height (and U) decreases, while its speed (and K) increases by exactly the same amount.

Conservation of Mechanical Energy

- It Cannot Be Created or Destroyed: This is the Law of Conservation of Energy. Energy can only be transformed from one type to another. The total amount of energy in the universe remains constant.

System	Type of Energy Conversion	Notes
Muscle contraction	Chemical → Mechanical	ATP breakdown, provides the energy to drives motion
Photosynthesis	Radiant (Light) → Chemical	Plants capture light energy and store it in the chemical bonds of sugars (like glucose).
Nerve signal	Chemical → Electrical	The movement of ions across a nerve cell membrane produces a voltage change to transmit signals.
Heart pumping	Chemical → Mechanical + Heat	Converts energy to pump blood throughout the body, with heat generated as a byproduct.
Respiration	Chemical → Thermal + Work	Work Releases the stored energy in glucose to provide heat for the body and energy for activities

- All living systems obey the same physical energy laws as machines, they just use them more efficiently and at microscopic scales.

Conservative Nonconservative Forces

- Most real systems have nonconservative forces like friction. In these cases, mechanical energy is not conserved.
- **The General Theorem:** The work done by all nonconservative forces is equal to the total change in mechanical energy.

$$W_{nc} = \Delta E + \Delta \mathrm{U} = (K_f + U_f) - (K_i + U_i)$$

W_{nc} : is the total work done by all nonconservative forces (most often friction).

- If W_{nc} : is **negative** (e.g., friction): Mechanical energy is lost or dissipated from the system (converted to thermal energy/sound). $\boldsymbol{E_f < E_i}$.
- If W_{nc} : is **positive** (e.g., a person pulling an object): Mechanical energy is added to the system (e.g., chemical energy from muscles is converted to mechanical energy). $\boldsymbol{E_f > E_i}$.
- This is the single most powerful energy equation. It works for all situations, whether nonconservative forces are present or not. (If $W_{nc} = 0$, it reduces to the conservation law).
- **Energy ,Work, and Power of the Body**

Under resting conditions about 25% of the body's energy is being used by the skeletal muscles and the heart ,19% is being used by the brain ,10% is used by the kidneys, and 27% is being used by the liver and spleen.

The body's basic energy (fuel) source is food.

The food is converted into molecules chemically.

· The body uses the food energy to operate its various organs:

Maintain constant temperature.

Do external work, for example, lifting.

A small percentage (5%) of the food energy is excreted in the feces and urine; any energy that is left over is stored as body fat.

The energy used to operate the organs eventually appears as body heat. Some of this heat is useful in maintaining the body at its normal temperature, but the rest must be disposed of.

- **Energy of food**

The human needs about 2400 Cal per day (for female about 2000 Cal/day) where:

1 (food) Calories = 1000 cal where 1 cal = 4.18J

Energy content per gram:

1 gram of fat gives 9 Calories.

1 gram of carbohydrates and sugars gives 4 Calories.

1 gram of proteins gives 4.3 Calories.

1 gram of Alcohol gives 7 Calories.

Example

One chocolate piece weights 100 g) 5 g of fat ,5 g of protein ,75 g of carbohydrates ,15 g of vitamins and water and others) how much calories in this piece?

- **Conservation of Energy in the Body**

Conservation of energy in the body can be written as a simple equation.

- [Change in stored energy] = Heat lost from the body + Work done

in the body (food energy, fat, and body heat)[

There are continuous energy changes in the body both when it is doing work and when it is not.

The first law of thermodynamic equation is:

$$\Delta U = \Delta Q + \Delta \mathrm{W}$$

ΔU is the change in stored energy, ΔQ is the heat lost or gained, $\Delta \mathrm{W}$ is the work done by the body in some interval of time.

A body doing no work $\Delta \mathrm{W} = 0$, and at a constant temperature loses heat to its surroundings, and ΔQ is negative.

ΔU :is also negative, indicating a decrease in stored energy.

The change of $\Delta U,\ \Delta Q,$ and $\Delta \mathrm{W}$ in a short interval of time $\Delta \mathrm{t}$

$$\frac{\Delta U}{\Delta t} = \frac{\Delta Q}{\Delta t} + \frac{\Delta w}{\Delta t}$$

$\frac{\Delta U}{\Delta t}$ is the rate of change of stored energy , $\frac{\Delta Q}{\Delta t}$ is rate of change of heat loss or gain, $\frac{\Delta w}{\Delta t}$ is the rate of doing work, that is mechanical power.

- The unit of energy in SI unit is Joule.

The physiological unit of food energy is Kilocalories.

The unit of heat production = Kcal/minute.

$$1\ Kcal\ =\ 4184\ J.$$

$$he\ body\ surface\ area. Met:\ is\ the\ rate\ of\ energy\ consumption\ of\ the\ body.$$

$$Power\ =\ Joule/second\ =\ Watts.$$

$$1\ Kcal\ =\ 4184\ J. 1\ Kcal\ =\ 4184\ J.$$

- **Metabolic rate (MR)**

A typical man has surface area 1.85 m^2A typical woman has about 1.4 m^2 of surface area.

1 met = 50 Kcal / hour per m^2 = 58 watts/m^2.

1 met = 92 kcal / hr.

1 met = 1.7 watts.

Metabolic rate: is defined as the rate of oxidation.

- In oxidation process within the body, heat is released as energy of metabolism.

Basal Metabolic Rate (BMR): is the lowest rate of energy consumption, or is defined as the amount of energy needed to perform minimal body functions (such as breathing and pumping the blood through the arteries) under resting conditions.

The energy used for basal metabolism becomes heat which is primarily dissipated from the skin, so the basal rate is related to the surface area and the mass of the body.

The metabolic rate depends on the temperature of the body. If the body temperature changes by 1°C ,there is a change of about 10 %in the metabolic rate.

❖ **Example**

In oxidation of glucose, heat energy is released:

$$C_6H_{12}O_2 \rightarrow o_2 = 6H_2o + 6CO_2 = 686Kcal$$

1 (mole) + 6 (mole) → $6\ (mole) + 6\ (mole) + heat\ energy$

1800 gm + 192 gm → 108 gm + 264 gm + 686 Kcal

Energy released per gm of glucose = 686/180 = 3.8 Kcal

Energy released per liter of O_2 used 686/6*22.4= 5.1 Kcal/liters

Liters of O_2 used per gm of fuel = 22.4*6/180=0.75 liters/gm

Liters of CO_2 produced per gm of fuel = 6*22.4/180=075 liters/gm.